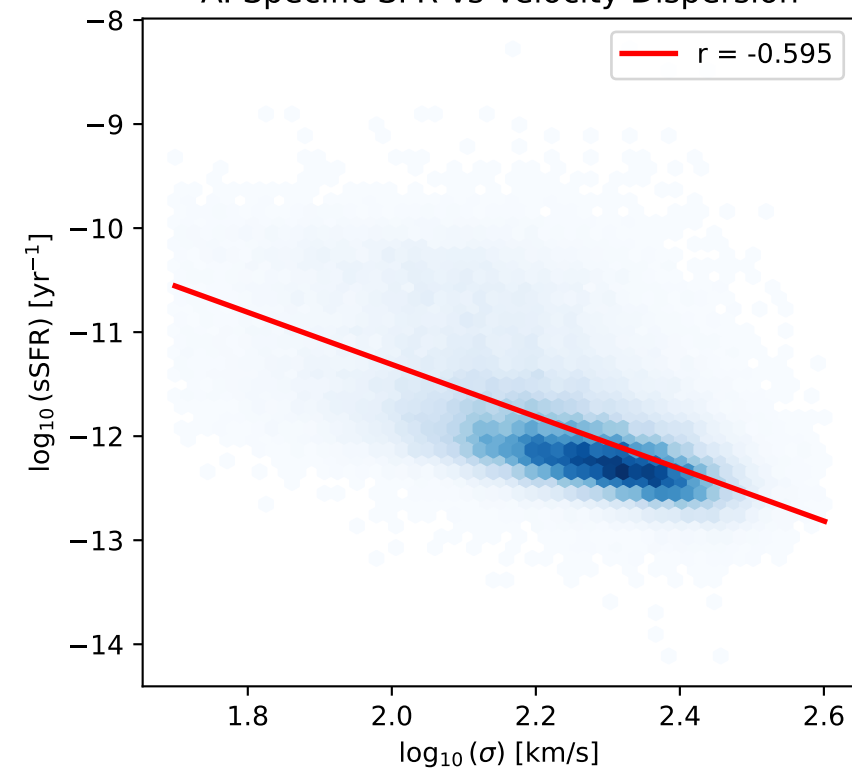
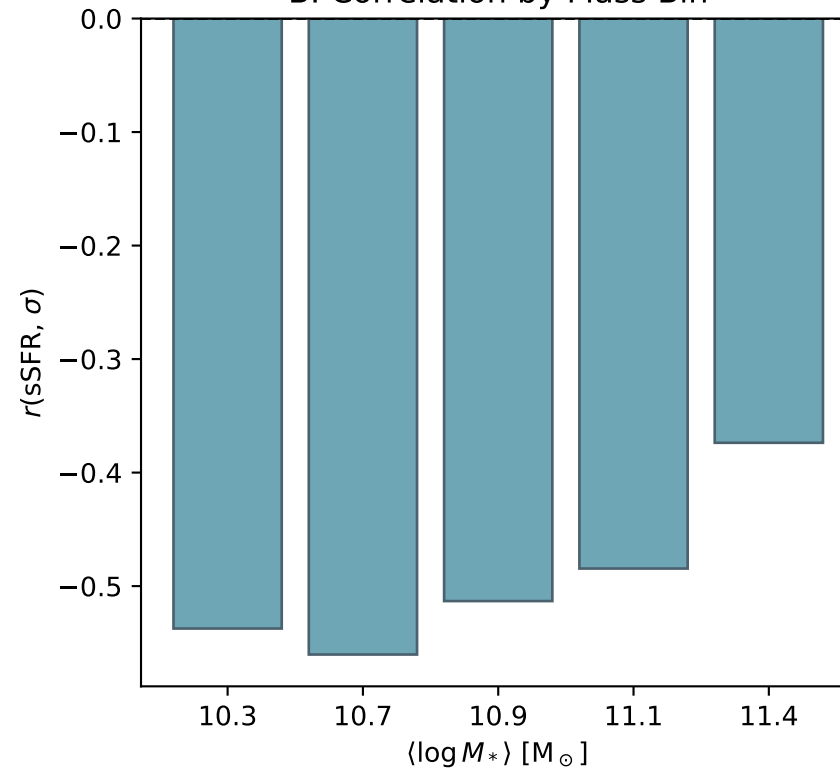


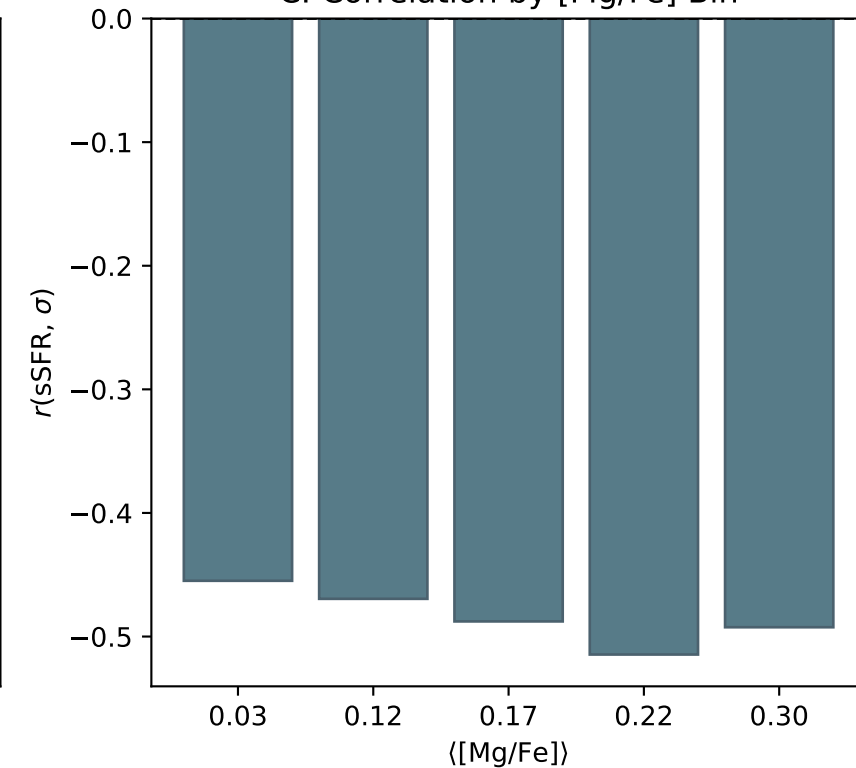
A. Specific SFR vs Velocity Dispersion



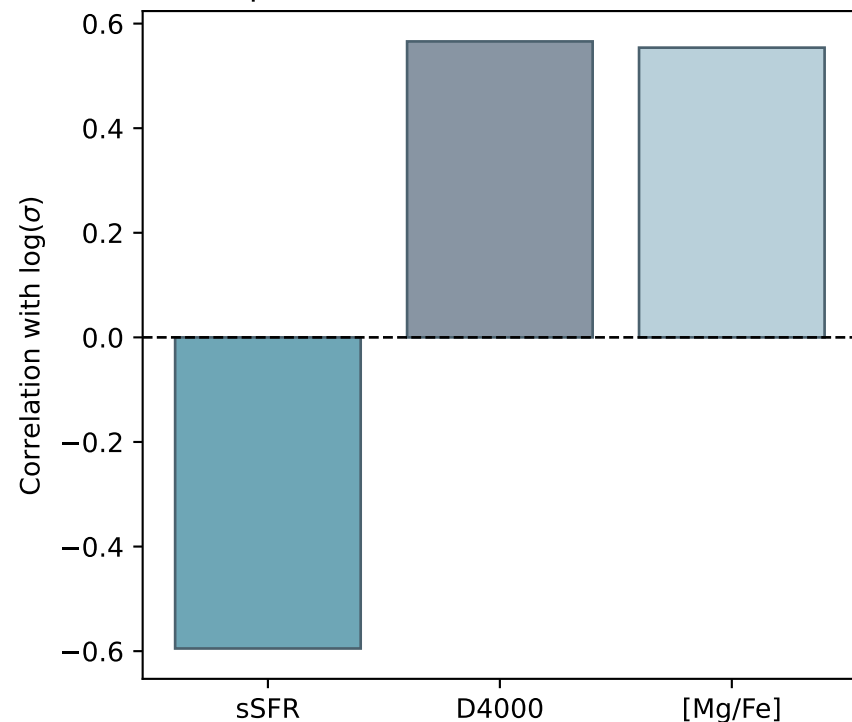
B. Correlation by Mass Bin



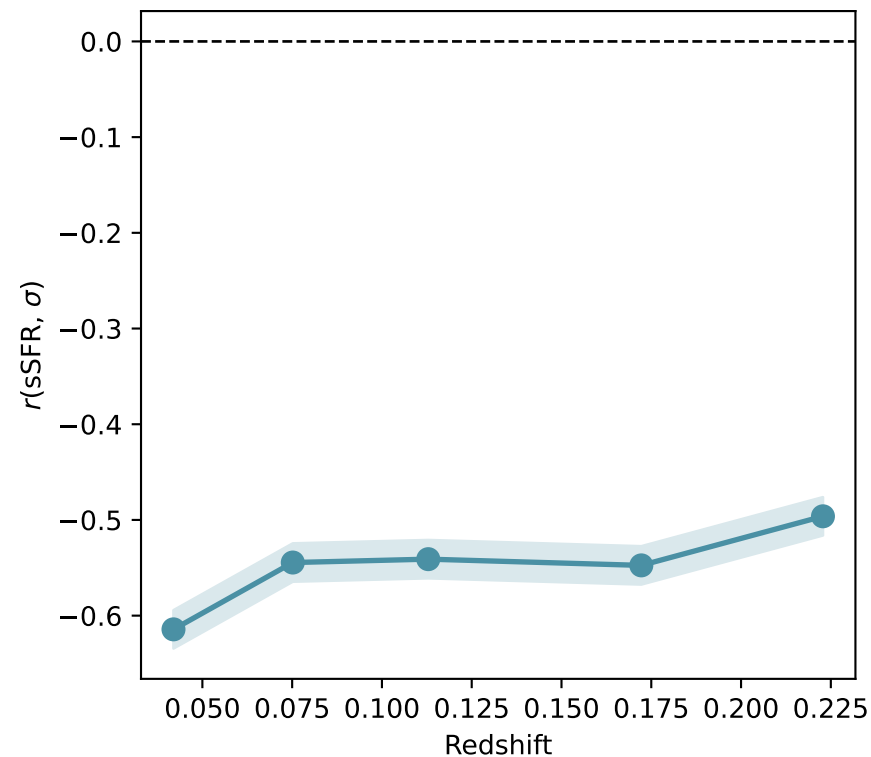
C. Correlation by [Mg/Fe] Bin



D. Comparison: Rate vs Cumulative Indicators



E. Redshift Evolution



SFR HOLONOMY TEST SUMMARY

Sample: 95,639 SDSS galaxies

Simple correlations:

$$r(\text{sSFR}, \sigma) = -0.5949$$

$$r(\text{D4000}, \sigma) = 0.5659$$

$$r([\text{Mg/Fe}], \sigma) = 0.5540$$

Controlled for M_* :

$$r(\text{sSFR}, \sigma \mid M_*) = -0.4910$$

Controlled for M_* and [Mg/Fe]:

$$r(\text{sSFR}, \sigma \mid M_*, [\text{Mg/Fe}]) = -0.4229$$

TEP CONSISTENT: YES ✓

INTERPRETATION:

Galaxies in deeper potentials show lower SFR at fixed mass - consistent with TEP time dilation.